

OEM MAINTENANCE & OPERATING MANUAL

Standard Operating Specification for Automated High-Speed Line

DOC ID: OEM-MAN-L2P-2025-V2

EQUIPMENT: LINE-2-PACKAGING

1. SYSTEM OVERVIEW & BASELINE OPERATIONAL PARAMETERS

LINE-2-PACKAGING operates under variable mechanical stress profiles governed by active SKU characteristics. Operational telemetry thresholds—specifically Vibration and Bearing Temperature—must be evaluated strictly against active IT production batch contexts.

Parameter	SKU-100 (High-Speed)	SKU-500 (Mid-Density)	SKU-899 (Heavy-Duty)
Nominal Speed (PPM)	450 units/min	320 units/min	180 units/min
Target OEE Baseline	88.0%	85.0%	80.0%
Normal Vibration Range	0.8 – 1.8 mm/s	1.2 – 2.2 mm/s	2.0 – 3.2 mm/s
Max Continuous Vibration	2.5 mm/s	3.0 mm/s	4.2 mm/s
Normal Bearing Temp	45°C – 62°C	50°C – 68°C	55°C – 75°C
Thermal Warning Cutoff	70°C	75°C	82°C

OPERATIONAL CONTEXT NOTE

Running **SKU-899** induces high mechanical torque on primary drive shafts. Vibration levels up to **3.2 mm/s** are baseline-normal for SKU-899, but constitute a **CRITICAL ANOMALY** if observed during a **SKU-100** run.

2. DIAGNOSTICS & ROOT CAUSE MATRIX

ERROR E-104

Main Drive Shaft Micro-Stoppage & Speed Degradation

Symptoms: OEE drop > 12% driven by performance loss; periodic vibration spikes (> 2.8 mm/s) on drive bearing B-12.

Root Causes:

- Poly-V Belt Slippage:** Occurs during long runs of SKU-500/899 under sustained torque load.
- Drive Roller Debris:** Resin accumulation from SKU-500 adhesive backing.

Corrective Action Procedure:

- Lockout/Tagout (LOTO) LINE-2-PACKAGING main electrical panel.
- Inspect Poly-V belt tension with sonic tension meter (Target: 140 N ± 5 N).
- Clean drive rollers using isopropyl alcohol (IPA) solution.
- Retighten tensioner bolt B-4 using a torque wrench set to 45 Nm.

ERROR E-208

Bearing Overheating & Thermal Runaway

Symptoms: Bearing Temperature on Motor Assembly M-2 exceeds 75°C during SKU-100 runs; accompanying high-frequency vibration (> 3.5 mm/s).

Root Causes:

- Grease Degradation:** High rotational speeds on SKU-100 cause thermal shear of Mobilith SHC 220 grease.
- Shaft Misalignment:** Axial misalignment exceeding 0.05 mm following SKU changovers.

Symptoms: High reject rate (Quality OEE drop), micro-stoppages under 30 seconds occurring every 10–15 minutes.

Root Causes:

- **Pneumatic Pressure Fluctuations:** Pressure drop below 5.8 bar during peak line throughput.
- **Thermal Mismatch:** Sealing jaw heater block operating off-spec during SKU-899 high-temp sealing cycle.

Corrective Action Procedure:

1. Verify main pneumatic regulator output is stable at 6.2 bar.
2. Re-calibrate PID temperature controller for Heating Element H-1 to 185°C for SKU-899.
3. Inspect seal jaw Teflon coating for wear; replace wear strips if pitting exceeds 0.2 mm.

3. MAINTENANCE PROTOCOLS & INSPECTION SCHEDULE

Weekly Preventive Maintenance (PM-W)

Perform the following physical inspections every 168 operating hours:

1. Clean optical sensors S-1 through S-8 with cleanroom-grade microfiber swabs.
2. Record baseline vibration readings across all 4 axis channels on Motor M-1 and M-2 using the portable analyzer.
3. Inspect pneumatic lines for micro-leaks or pressure drops.

Monthly Preventive Maintenance (PM-M)

Perform the following comprehensive checks every 720 operating hours:

1. Check alignment of primary drive sprockets and verify chain slack (10–15 mm deflection allowance).
2. Lubricate drive shaft bearings with 12g to 15g of Mobilith SHC 220 grease.
3. Inspect electrical connection torque settings on main drive VFD modules (3.5 Nm required).

4. AUTOMATION & AGENT RESOLUTION METADATA

The structured metadata block below is embedded for automatic indexing and vector retrieval via Snowflake Cortex Search and multi-agent orchestration engines.

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